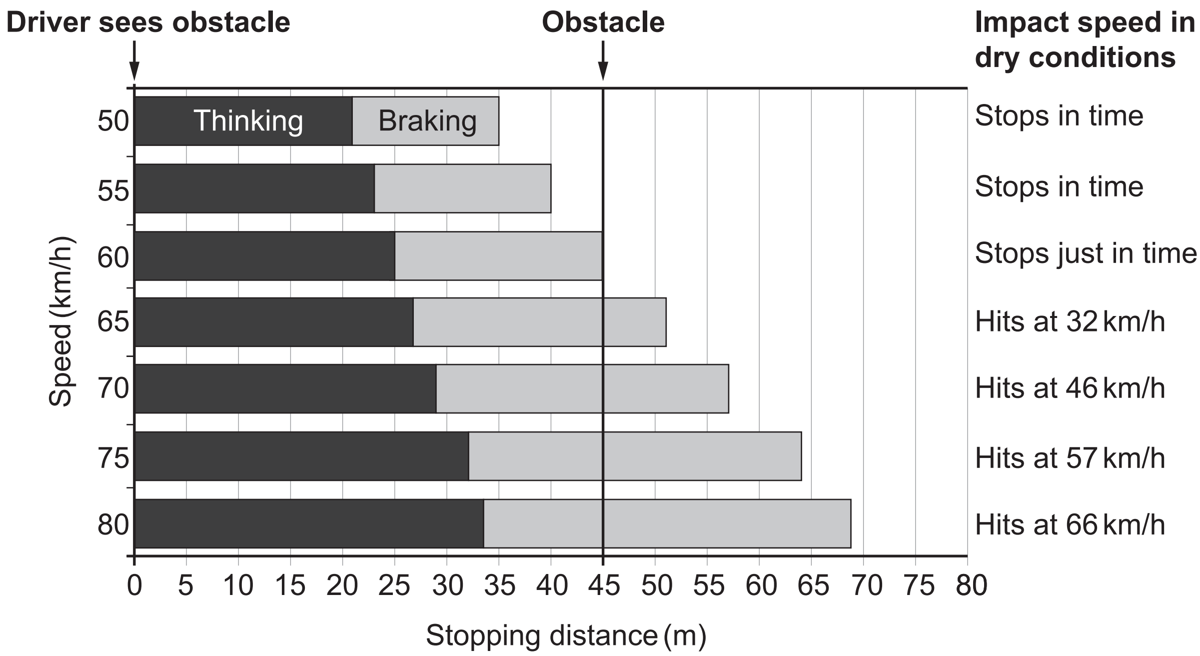
Examiner only

7. The chart below is used by traffic collision investigators. It gives the thinking, braking and stopping distances of cars driven at different speeds by an alert driver on a dry road.

An alert driver notices an obstacle 45 m away on the road ahead. The position of this obstacle is represented by the dark vertical line. If there is a collision, the chart also shows the impact speed with the obstacle.



(a) State how the following information in the chart for a speed of 70 km/h would compare if the tyre treads on the car are worn below the legal limit. [3]

(i) Thinking distance

....

(ii) Braking distance

....

(iii) Impact speed

....

Examiner only

(b) Use the information opposite to answer the following questions about a car travelling at 60 km/h which **decelerates to a stop**.
(10 km/h = 2.8 m/s)

(i) **Complete** the following table. [4]

Initial speed (km/h)	Initial speed (m/s)	Thinking distance (m)	Braking distance (m)	Stopping distance (m)
60	....	....	....	....

(ii) Using the information in the table, calculate the **thinking time** of the driver. [3]

thinking time = ... s

(iii) Use the equation:

$$v^2 = u^2 + 2ax$$

and the information in the table to calculate the deceleration of the car. [3]

deceleration = ... m/s^2

(c) The chart shows that a car travelling faster than 60 km/h will not stop in time and therefore collides with the obstacle.

(i) Explain, in terms of Newton's 1st Law, how seat belts provide protection during a collision. [2]

....

....

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Examiner only

(ii) Explain, in terms of Newton's 3rd law, why the car and the obstacle are damaged during a collision. [2]

....

....

....

(iii) Explain, in terms of Newton's 2nd law, why cars have crumple zones. [2]

....

....

....

(d) The Royal Society for the Prevention of Accidents (RoSPA) says that "The majority of pedestrian casualties (in the UK) occur in built-up areas: 22 of the 26 child pedestrians and 264 of the 372 adult pedestrians who were killed in 2017, died on roads in built-up areas."
They strongly feel that reducing the speed limit in built-up areas from 50 km/h to 35 km/h would greatly improve these accident statistics.
Explain whether you agree with RoSPA. [3]

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....

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END OF PAPER